Mimic X Adapter — Sharp X68000 Keyboard / Mouse

USB-MIDI in from a phone, X68000 mini-DIN 7P keyboard port out. Keyboard and mouse share the one connector.

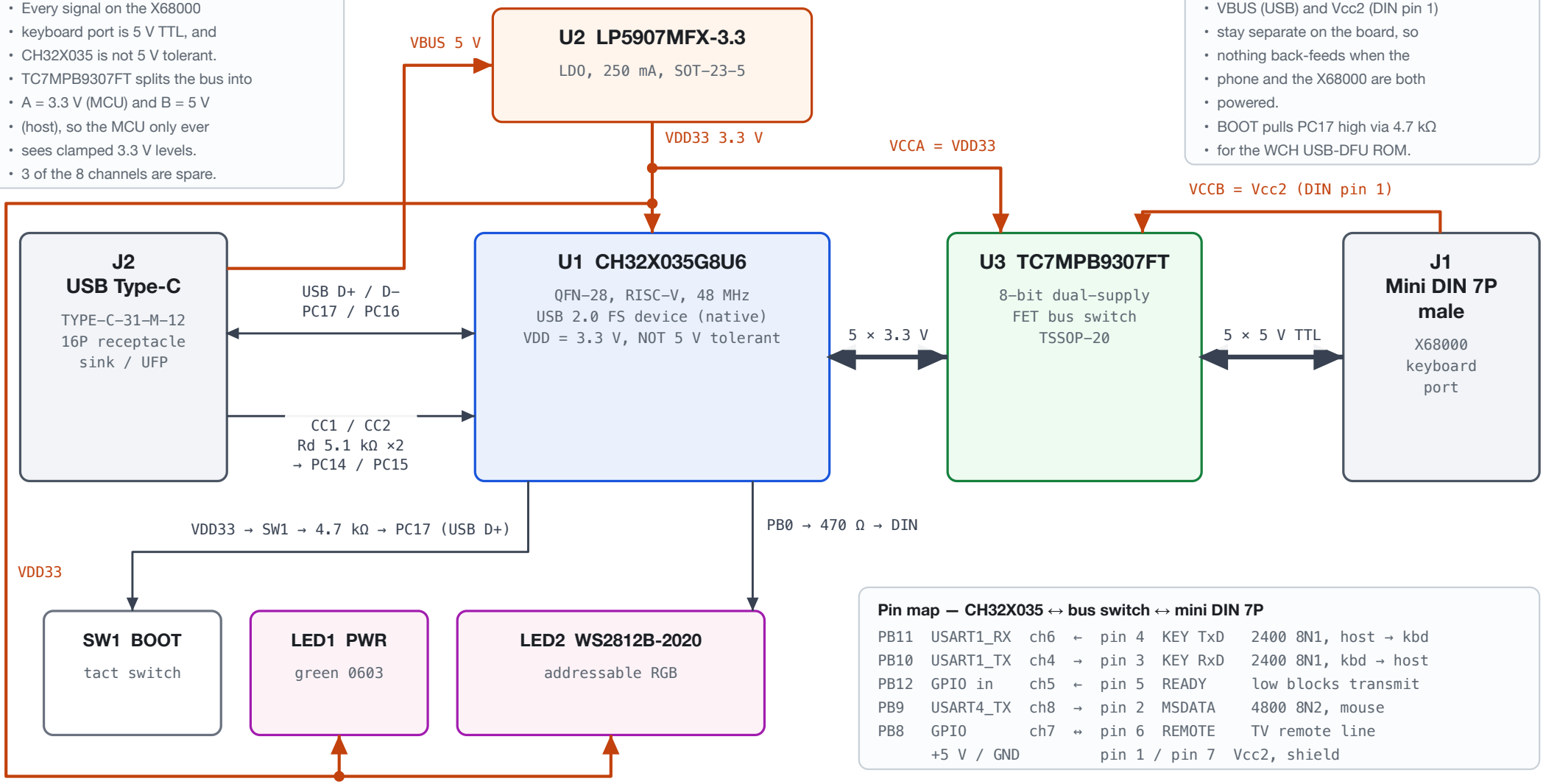
board rev 1.0.0 · CERN-OHL-S-2.0
USB VID:PID 0x1209:0x9210
github.com/kunichiko/MimicX-hardware

Why the bus switch

- Every signal on the X68000
- keyboard port is 5 V TTL, and
- CH32X035 is not 5 V tolerant.
- TC7MPB9307FT splits the bus into
- A = 3.3 V (MCU) and B = 5 V
- (host), so the MCU only ever
- sees clamped 3.3 V levels.
- 3 of the 8 channels are spare.

Power domains

- VBUS (USB) and Vcc2 (DIN pin 1)
- stay separate on the board, so
- nothing back-feeds when the
- phone and the X68000 are both
- powered.
- BOOT pulls PC17 high via 4.7 kΩ
- for the WCH USB-DFU ROM.



Pin map — CH32X035 ↔ bus switch ↔ mini DIN 7P

PB11	USART1_RX	ch6	←	pin 4 KEY TxD	2400 8N1, host → kbd
PB10	USART1_TX	ch4	→	pin 3 KEY RxD	2400 8N1, kbd → host
PB12	GPIO in	ch5	←	pin 5 READY	low blocks transmit
PB9	USART4_TX	ch8	→	pin 2 MSDATA	4800 8N2, mouse
PB8	GPIO	ch7	↔	pin 6 REMOTE	TV remote line
	+5 V / GND			pin 1 / pin 7	Vcc2, shield

power rail — signal — multi-bit bus

Every net shown here is declared in the .ato source; this drawing is generated from it by hand, not extracted.